

1 Introduction

We study the interpolated stochastic heat and wave equation (ISHWE) for spatial dimension on $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & \beta \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & \beta \in (1, 2). \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

2 Preliminaries

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write (assuming $\beta \in (1, 2)$)

$$u(t, x) = J_0(t, x) + I(t, x)$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

To establish the existence and uniqueness of random field solutions to (1.1), we will need to assume Dalang's condition:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.1) \quad \boxed{\text{C:Dalang}}$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta. \quad (2.2) \quad \boxed{\text{E:DalangEqu}}$$

3 1-d Case

In this section we study (1.1) for $d = 1$.

3.1 Globally Lipschitz Coefficient

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|,$$

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ . We assume that $L(\sigma) > 0$. We now state an existence and uniqueness result proven in [2].

alityGenLip

Theorem 3.1. *Under (2.1), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that*

$$\hat{C}_t := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (3.1)$$

Moreover, the following statements are true:

1. $(t, x) \mapsto u(t, x)$ is $L^P(\Omega)$ -continuous for all $p \geq 2$,

2. For all even integers $p \geq 2$, all $t \geq 0$ and $x, y \in \mathbb{R}^d$,

$$\|u(t, x)\|_p^2 \leq 2J_0^2(t, x) + \left[\zeta^2 + 2\hat{C}_t^2\right] \exp(C\lambda^{2/(1-\sigma)}t). \quad (3.2)$$

1and2normYZ

Lemma 3.2. [2, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha},$$

for all $t > 0$, where

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}} Z(t, x) dx = t^{\lceil \beta \rceil - 1} = t.$$

Remark 3.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 3.2 gives us that $\int_{\mathbb{R}} G^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (3.3)$$

E:SUIdentifi

I believe that Proposition 3.4 below can be proven for a more general dimension d .

Proposition 3.4. [4, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$. Then there exists a universal constant $K := K(\alpha, \beta, \gamma) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2}\right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (3.4)$$

E:Nbdd

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)}\right]^p, \quad t \in [0, T]. \quad (3.5)$$

E:Pnormtbdd

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 3.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + \frac{\partial}{\partial t}(Z(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (3.3), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \end{aligned} \quad (3.6) \quad \boxed{\text{E:Jbdd}}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 3.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y).$$

Now under Dalang's condition [2], $(d/\alpha - 2)\beta - 2\gamma < -1$, the first integral above can be calculated as

$$\int_0^t (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha} ds = \frac{\alpha t^{(2-d/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d}.$$

By applying Lemma 3.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y) = C_\# \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{\alpha t^{(2-d/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_\# \alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} t^{(2-d/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{-2a(t-s)} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\|I(t, x)\|_p^2 e^{-2at} \leq \frac{8pc^2(\sigma)C_\# \alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} t^{(2-d/\alpha)\beta+2\gamma-1} e^{-2at}$$

$$\begin{aligned}
& + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#}\int_0^\infty du e^{-2a(u)}(u)^{2(\beta+\gamma-1)-\beta d/\alpha}. \\
& = \frac{8pc^2(\sigma)C_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}t^{(2-d/\alpha)\beta+2\gamma-1}e^{-2at} \\
& \quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#}(2a)^{1+(d/\alpha-2)\beta-2\gamma}\Gamma[(2-d/\alpha)\beta-1+2\gamma]
\end{aligned}$$

where the last line is true if Dalang's condition [2] holds. If we now apply (3.3) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$\begin{aligned}
N_{a,p}(I) & \leq c(\sigma)\sqrt{\frac{8pC_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}}[(2-d/\alpha)\beta/2+\gamma-1/2]^{(2-d/\alpha)\beta/2+\gamma-1/2}(ea)^{1/2-\gamma-(2-d/\alpha)\beta/2} \\
& \quad + L(\sigma)N_{a,p}(u)\sqrt{8pC_{\#}(2a)^{1+(d/\alpha-2)\beta-2\gamma}\Gamma[(2-d/\alpha)\beta-1+2\gamma]}. \tag{3.7}
\end{aligned}$$

E:SoleMil3..

To simplify things, we now let

$$\begin{aligned}
K := \max \Bigg\{ & \sqrt{\frac{8C_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}} \left(\frac{(2-d/\alpha)\beta/2+\gamma-1/2}{e} \right)^{(2-d/\alpha)\beta/2+\gamma-1/2}, \\
& \left(\frac{1}{2} \right)^{(2-d/\alpha)\beta/2+\gamma-1/2} \sqrt{8C_{\#}\Gamma[(2-d/\alpha)\beta-1+2\gamma]} \Bigg\}
\end{aligned}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}}(c(\sigma) + L(\sigma)N_{a,p}(u)). \tag{3.8}$$

E:Ibdd

Now by combining (3.6) and (3.8), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|u_0\|_\infty & \beta \in (0,1] \\ (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1,2). \end{cases} \tag{3.9}$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}}N_{a,p}(u). \tag{3.10}$$

E:Nbddgen

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (3.7) and (3.10) recover [4, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (3.10) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|u_0\|_\infty & \beta \in (0,1] \\ (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1,2) \end{cases}$$

$$+ \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (3.4). Note that (3.5) follows immediately from (3.4). $\square$

We will need to recall the following result proved in [2] in order to prove Proposition 3.7 below.

P:YInc

Proposition 3.5. [1, Proposition 2.1 and 2.2] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.2) holds. Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

P:YIncCase1

1. *Suppose that $0 < \theta < (\Theta - 1) \wedge 2$ where Θ is given in (2.2), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq \lceil \beta \rceil - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right),$$

where

$$\rho := 2(\beta + \gamma) - 1 - \beta d / \alpha. \quad (3.11)$$

D:rho

P:YIncCase2

2. *If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (2.2) holds then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 3.6. In Proposition 3.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 3.7. *Suppose that $u_0 = 1$ and that $v_0 = 0$ so that $u(t, x) = 1 + I(t, x)$. Let $a > 0$, $\rho = 2(\beta + \gamma) - 1 - \beta d / \alpha$ and suppose that $0 < \theta < (\Theta - 1) \wedge 2$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$,*

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (3.12)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{pL(\sigma)},$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^\rho + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (3.13)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 3.4, to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}.\end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right].\end{aligned}$$

We now apply Proposition 3.5 to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right],\end{aligned}$$

where $C(\theta, T)$ is as in Proposition 3.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (3.12). It is clear that (3.13) follows directly from (3.12) and Proposition 3.4. $\square$

The holder continuity in the next proposition needs to be checked

P:Holder

Proposition 3.8. Suppose that u_0 and v_0 are as on Proposition 3.7 above. Then u has a version, still denoted by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 3.4, then we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

Proof. Proposition 3.7 implies that

$$\|u(t, x) - u(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (3.14)$$

By Kolmogorov's continuity theorem, there exists a version of u , which we denote by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|u_0\|_\infty^p + |u(t, x) - u_0(x)|^p) \\ &\leq 2^{p-1} \left(\|u_0\|_\infty^p + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - u_0(x)|^p) \right) \\ &= 2^{p-1} \left(1 + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) \right) \end{aligned}$$

Due to the continuity of u on the compact set $[0, T] \times [-R, R]$, there exists a point $(t_0, x_0) \in [0, T] \times [-R, R]$ such that

$$\sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) = |u(t_0, x_0) - 1|^p.$$

Thus, for any $(t, x) \in [0, T] \times [-R, R]$,

$$|u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p)$$

which in return implies that

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p).$$

Now by taking the expectation of both sides and applying Proposition 3.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] &\leq 2^{p-1} + 2^{p-1} C(p, \theta, T)^p \left((2T)^{\rho/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right],$$

then by Proposition 3.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 3.9. The Hölder continuity in time proven above in Proposition 3.8 recovers the result proven by Foondun and Nane in [3, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

3.2 Existence and Uniqueness of a Global Solution with Superlinear Diffusion Coefficient

We now impose the following condition on σ : As $|x|, |z| \rightarrow \infty$,

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (3.15)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (3.16)$$

where $\sigma_1 = c(\sigma) = \sigma(0)$.

Remark 3.10. Because of the absence of the drift coefficient in (1.1), we may drop [4, Assumption C1].

Theorem 3.11. Consider the initial conditions $u_0 = 1$ and $v_0 = 0$ and suppose that σ satisfies (3.15). Then for any $M > 0$, there exists a random field solution to (1.1) in $[-M, M]$ denoted as $(u(t, x) : (t, x) \in [0, T] \times [-M, M])$. This solution is unique and satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, \text{ a.s.} \quad (3.17)$$

Proof. We first solve a truncated version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where

$$\sigma_N(x) = \sigma(x) \mathbf{1}_{\{|x| \leq N\}} + \sigma(N) \mathbf{1}_{\{x > N\}} + \sigma(-N) \mathbf{1}_{\{x \leq -N\}}.$$

Note that since σ satisfies (3.15), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 3.1 and see that there exists a unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 3.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 3.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 3.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.18)$$

where we recall that

$$\mathcal{M}_1 = \sqrt{p} \sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.19)$$

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (3.15). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.20)$$

The uniqueness of the solution and the [local property of the stochastic integral](#) imply that $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T) \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T) \times \mathbb{R}.$$

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in [Dalang and Sanz-Sole's textbook that is not yet published](#).

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right). \end{aligned}$$

An application of [\(3.18\)](#) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using [\(3.19\)](#) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow \infty, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

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